

Eva Dimitrova

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EDUCATION

University of Maryland, College Park

Bachelor of Science in Computer Engineering, GPA: 3.6/4.0

College Park, MD

Expected May 2028

WORK AND LEADERSHIP EXPERIENCE

Software Engineering Intern (Incoming)

Liberty Mutual Insurance

May 2026 – August 2026

Portsmouth, NH

- Working with Java, Spring Boot, AWS, and REST APIs to build scalable backend services

Computer Engineering Intern

Brown Deer Technology

May 2025 – Aug 2025

Bel Air, MD

- Optimized a C-based libconfig parser on Linux, improving automation and testing efficiency by 50%
- Configured and validated the seL4 microkernel on RISC-V architecture, debugging low-level system behavior to ensure accurate and reliable system operation for scalable deployment

Research Intern

Maryland Information and Network Dynamics Lab

May 2025 – August 2025

College Park, MD

- Developed customized anomaly detection algorithms for building energy sensor data across 30+ meters, reducing false positives by 60% and enabling precise time-bounded anomaly identification for downstream analysis.
- Built a scalable visualization system that surfaces detected anomalies directly on time-series sensor data, eliminating manual graph inspection and supporting validation workflows for 15+ researchers.

ECE Department Peer Mentor

A. James Clark School of Engineering

September 2025 – Present

College Park, MD

- Mentor 50+ students in debugging strategies, systems concepts, and project development

PROJECTS

YouTube Semantic Video Search System | *Python, FastAPI, FAISS, JavaScript*

January 2026

- Built semantic search system over YouTube liked videos using embeddings and FAISS for similarity-based retrieval
- Developed FastAPI backend with REST APIs for efficient search queries and low-latency response handling
- Integrated YouTube Data API with OAuth2 authentication to automate video metadata collection

Student Performance Predictor Model | *Python, React, Flask, PyTorch, XGBoost*

August 2025

- Engineered end-to-end ML pipeline including preprocessing, feature engineering, model training, and deployment
- Processed 30+ features through cleaning, encoding, and normalization to improve model performance
- Trained XGBoost model achieving 91% accuracy to predict student outcomes
- Built interactive React + Flask interface to visualize predictions

Social Learning Platform | *React, Supabase, SQL*

January 2025

- Developed a full-stack social learning platform for active study groups using React and Supabase
- Designed SQL-backed data models and implemented real-time updates to support dynamic group activity

ACTIVITIES

Capital One Fintech Future Hackathon

May 2025

Full-Stack Engineer

- Built full-stack budgeting application integrating Plaid API for real-time financial data and goal tracking
- Developed front-end features and backend integrations supporting peer-based savings incentives

App Development Club

February 2025 – May 2025

Machine Learning Engineer

- Contributed to a RAG-based financial document analysis tool using LangChain and Amazon Bedrock for natural language querying over investment PDFs
- Implemented PDF parsing and tabular data extraction to structure financial reports for further analysis

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, OCaml, Rust, Postgres/SQL, R

Technologies: Git/GitHub, Linux, FastAPI, PyTorch, Flask, Node.js, React, Figma, AWS, XGBoost, FAISS